TECHNICAL DOCUMENTATION

CareerTrack – Backend REST API Specification

1. Overview & Architecture

The CareerTrack Backend REST API provides a secure, decoupled service built on Node.js and Express.js. It implements a layered controller-service-route-model architecture with strict server-side validation, centralized error handling, and robust security headers.

- **Base URL:** `http://localhost:5000/api` (or deployed base URL)
- **Data Format:** JSON (`application/json`) for standard requests; `multipart/form-data` for candidate application submissions with résumé uploads.
- **Authentication:** Token-based authentication using JSON Web Tokens (JWT). Admin protected endpoints require an `Authorization` header formatted as:

  ```
  Authorization: Bearer <your_jwt_token>
  ```

- **Rate Limiting:**
  - Global API: 300 requests per 15 minutes per IP.
  - Auth Login: 20 attempts per 15 minutes to prevent credential brute-forcing.

2. API Endpoint Directory

Method	Endpoint	Access	Purpose
GET	`/health`	Public	Verify backend service availability
GET	`/jobs`	Public	Browse published jobs with search, multi-filters & pagination
GET	`/jobs/:id`	Public	Retrieve single published job details
POST	`/applications/apply/:jobId`	Public	Submit candidate application & résumé without login
POST	`/auth/login`	Public	Authenticate administrator & issue JWT token
GET	`/auth/me`	Protected (Admin)	Retrieve authenticated administrator profile
GET	`/jobs/admin/all`	Protected (Admin)	List all jobs across all lifecycle statuses
GET	`/jobs/admin/analytics/summary`	Protected (Admin)	Dashboard telemetry, counts & pipeline distribution
POST	`/jobs`	Protected (Admin)	Create a new job opening
PUT	`/jobs/:id`	Protected (Admin)	Update an existing job opening
PATCH	`/jobs/:id/status`	Protected (Admin)	Quick status change (Publish, Unpublish, Close, Draft)

Method	Endpoint	Access	Purpose
DELETE	`/jobs/:id`	Protected (Admin)	Delete job posting and associated applications
GET	`/applications/admin/all`	Protected (Admin)	Filter, search & paginate candidate applications
GET	`/applications/admin/:id`	Protected (Admin)	Retrieve full candidate profile, qualifications & notes
PATCH	`/applications/admin/:id/status`	Protected (Admin)	Update recruitment stage (`Shortlisted` , `Selected` , etc.)
POST	`/applications/admin/:id/notes`	Protected (Admin)	Add internal recruiter evaluation note with timestamp
GET	`/applications/admin/:id/resume`	Protected (Admin)	Securely stream and download candidate's uploaded résumé
GET	`/applications/admin/export/csv`	Protected (Admin)	Export application records to spreadsheet CSV format

3. Public Endpoints Specification

3.1. Health Check

Endpoint: `GET /api/health`

Response:

```
{
  "success": true,
  "status": "online",
  "timestamp": "2026-10-29T05:34:05.595Z",
  "service": "CareerTrack REST API"
}
```

3.2. Get Published Active Jobs

Endpoint: `GET /api/jobs` | **Access:** Public

Query Parameters:

- `search` (string): Keywords matching title, company, skills, or location.
- `location` (string): Specific city or region.
- `employmentType` (string): `Full-time` , `Part-time` , `Contract` , `Internship` .
- `experienceLevel` (string): `Entry-level` , `Mid-level` , `Senior-level` , `Lead` .
- `isRemote` (boolean): `true` to filter remote roles.
- `sortBy` (string): `newest` (default), `oldest` , `salaryHigh` , `deadline` .
- `page` (integer): Page number (default: 1).
- `limit` (integer): Records per page (default: 10).

Success Response (200 OK):


```
{
  "success": true,
  "count": 6,
  "total": 6,
  "totalPages": 1,
  "currentPage": 1,
  "jobs": [
    {
      "_id": "67215f9b4c09d31e9a2b8401",
      "title": "Senior Full-Stack Engineer (MERN)",
      "company": "InnovateTech Solutions",
      "location": "San Francisco, CA",
      "isRemote": true,
      "employmentType": "Full-time",
      "experienceLevel": "Senior-level",
      "department": "Engineering",
      "salaryMin": 140000,
      "salaryMax": 185000,
      "salaryCurrency": "USD",
      "isSalaryDisclosed": true,
      "skills": ["React.js", "Node.js", "Express.js", "MongoDB"],
      "status": "Published",
      "postingDate": "2026-10-29T05:00:00.000Z",
      "applicationDeadline": "2026-11-28T05:00:00.000Z"
    }
  ]
}
```


3.3. Get Single Job Details

Endpoint: `GET /api/jobs/:id` | **Access:** Public

Success Response (200 OK):


```
{
  "success": true,
  "job": {
    "_id": "67215f9b4c09d31e9a2b8401",
    "title": "Senior Full-Stack Engineer (MERN)",
    "company": "InnovateTech Solutions",
    "description": "InnovateTech is seeking an experienced Senior Full-Stack Engineer...",
    "responsibilities": ["Architect, build and maintain scalable full-stack applications"],
    "qualifications": ["5+ years of production experience in JavaScript/TypeScript"],
    "benefits": ["Competitive equity package", "Unlimited PTO"],
    "skills": ["React.js", "Node.js", "MongoDB"],
    "isExpired": false,
    "isAcceptingApplications": true
  }
}
```


Error Response (403 Forbidden):

```
{
  "success": false,
  "message": "This job posting is currently not active or available to the public."
}
```

3.4. Candidate Application Submission

Endpoint: `POST /api/applications/apply/:jobId` | **Access:** Public (No authentication required)

Content-Type: `multipart/form-data`

Form Body Fields:

- `candidateName` (string, required): Legal full name.
- `email` (string, required): Valid email address.
- `phone` (string, required): Contact number.
- `education` (string, required): Highest education degree.
- `experienceYears` (number, required): Years of relevant experience.
- `currentRole` (string, optional): Current role & company.
- `technicalSkills` (string, optional): Comma-separated skill list.
- `coverLetter` (string, optional): Cover letter notes.
- `resume` (file, required): File stream in `.pdf`, `.docx`, or `.doc` (Max 5 MB).

Success Response (201 Created):

```
{
  "success": true,
  "message": "Your application has been successfully submitted! Our recruiting team will rev
iew your qualifications.",
  "applicationReferenceId": "CT-2026-Q8X1A",
  "submittedAt": "2026-10-29T05:40:12.304Z"
}
```

Duplicate Prevention Response (409 Conflict):

```
{
  "success": false,
  "message": "Duplicate Submission: An application from 'alex.morgan.dev@example.com' has al
ready been registered for 'Senior Full-Stack Engineer (MERN)'."
}
```

Closed / Expired Job Rejection (400 Bad Request):

```
{
  "success": false,
  "message": "This job posting is currently unpublished, closed, or the application deadline has passed."
}
```

4. Administrator Protected Endpoints

4.1. Administrator Login

Endpoint: `POST /api/auth/login` | **Access:** Public

Request Body:

```
{
  "email": "admin@careertrack.com",
  "password": "AdminPassword123!"
}
```

Success Response (200 OK):

```
{
  "success": true,
  "message": "Authentication successful",
  "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "admin": {
    "id": "67215f9b4c09d31e9a2b83ff",
    "name": "Talent Acquisition Director",
    "email": "admin@careertrack.com",
    "role": "superadmin",
    "lastLogin": "2026-10-29T05:35:00.000Z"
  }
}
```

4.2. Dashboard Metrics & Telemetry

Endpoint: `GET /api/jobs/admin/analytics/summary` | **Headers:** `Authorization: Bearer <token>`

Success Response (200 OK):

```
{
  "success": true,
  "metrics": {
    "totalJobs": 7,
    "activeJobs": 6,
    "closedJobs": 1,
    "expiredJobs": 1,
    "totalApplications": 4,
    "statusBreakdown": {
      "Submitted": 1,
      "Under Review": 1,
      "Shortlisted": 1,
      "Interview Scheduled": 1,
      "Selected": 1,
      "Rejected": 0
    },
    "recentApplications": [...],
    "recentJobs": [...]
  }
}
```

4.3. Update Candidate Application Status

Endpoint: `PATCH /api/applications/admin/:id/status` | **Headers:** `Authorization: Bearer <token>`

Request Body:

```
{
  "status": "Interview Scheduled"
}
```

Success Response (200 OK):

```
{
  "success": true,
  "message": "Candidate application status updated to: Interview Scheduled",
  "application": {
    "_id": "67216a044c09d31e9a2b840e",
    "candidateName": "Alex Morgan",
    "status": "Interview Scheduled"
  }
}
```

4.4. Add Internal Recruiter Note

Endpoint: `POST /api/applications/admin/:id/notes` | **Headers:** `Authorization: Bearer <token>`

Request Body:

```
{
  "note": "Candidate demonstrated exceptional knowledge of Node.js event loop and Mongoose a
ggregation pipelines in screening round."
}
```

Success Response (201 Created):

```
{
  "success": true,
  "message": "Recruitment note added successfully",
  "notes": [
    {
      "_id": "67216c804c09d31e9a2b8430",
      "note": "Candidate demonstrated exceptional knowledge...",
      "authorName": "Talent Acquisition Director",
      "createdAt": "2026-10-29T05:52:00.000Z"
    }
  ]
}
```

4.5. Export Applications to CSV

Endpoint: `GET /api/applications/admin/export/csv` | **Headers:** `Authorization: Bearer <token>`

Query Parameters: `jobId` (optional), `status` (optional)

Response Headers:

- `Content-Type: text/csv`
- `Content-Disposition: attachment; filename="careertrack_applications_<timestamp>.csv"`

Response Body: CSV file content stream containing all candidate records, job titles, and recruitment statuses.